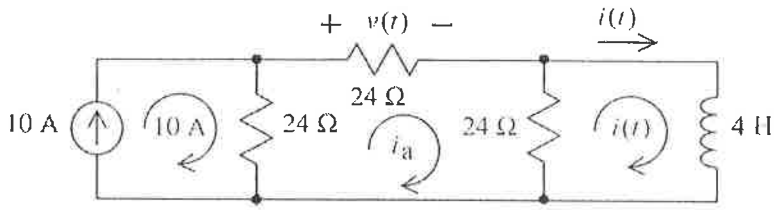
P7.5-13



i5pt

5pt.

We'll write and solve a mesh equation. Label the meshes as shown. Apply KVL to the center mesh to get

5pt.

$$24i_a + 24\left(i_a - i(t)\right) + 24\left(i_a - 10\right) = 0 \quad \Rightarrow \quad i_a = \frac{i(t)+10}{3} = 5 - e^{-4t} \text{ A for } t > 0$$

Then

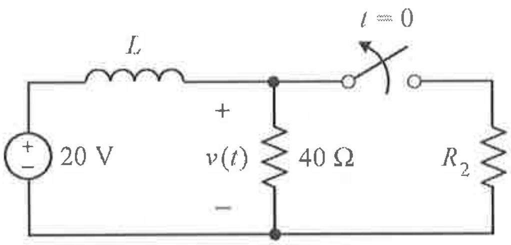
$$v(t) = 24i_a = 120 - 24e^{-4t} \text{ V for } t > 0$$

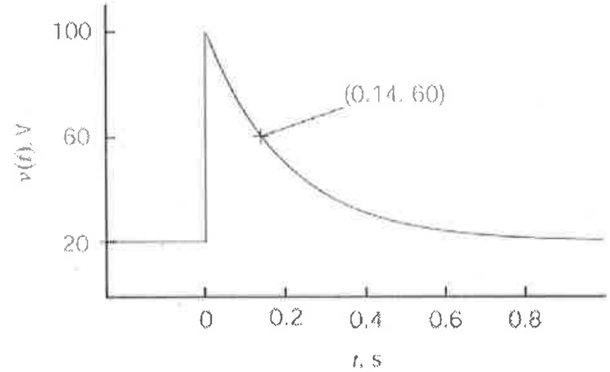
5pt.

(checked: LNAP 6/25/04)

15

P8.3-25





From the plot $D =$ $v(t)$ for $t < 0$ =20 V, $E + F = v(0+) = 100$ V and $E = \lim_{t\to\infty} v(t) = 20\text{ V}$. The point labeled on the plot indicates that $v(t) = 60$ V when $t =$ 0.14 s. Consequently 3

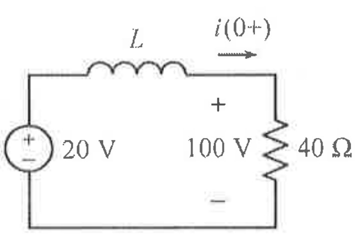
$$60 = 20 + 80\,e^{-a(0.14)} \quad \Rightarrow \quad a = \frac{\ln\left(\frac{60-20}{80}\right)}{-0.14} = 5\ \frac{1}{\text{s}}$$ 3

Then

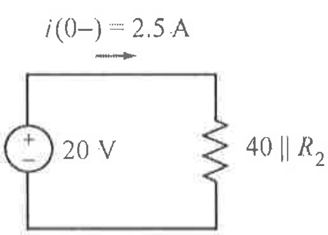
$$v(t) = \begin{cases} 20\text{ V} & \text{for } t \le 0 \\ 20 + 80\,e^{-5t}\text{ V} & \text{for } t \ge 0 \end{cases}$$

At $t = 0+$,

$$i(0+) = \frac{100}{40} = 2.5\text{ A}$$ 3



When $t < 0$, the circuit is at steady state so the inductor acts like a short circuit.



$$40 \,||\, R_2 = \frac{20}{2.5} = 8\ \Omega \Rightarrow \quad R_2 = 10\ \Omega$$ 3.

3

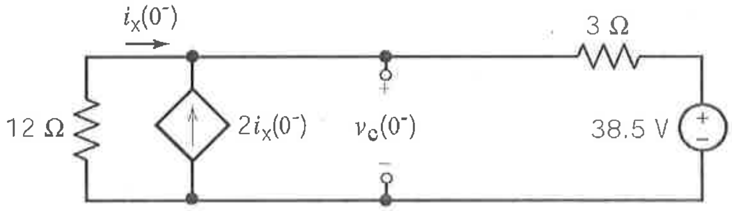
Next, the inductance can be determined using the time constant:

$$5 = a = \frac{1}{\tau} = \frac{40}{L} \quad \Rightarrow \quad L = \frac{40}{5} = 8\text{ H}$$

20pt

P8.7-1

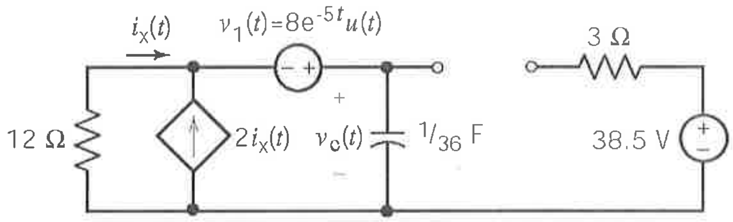
Assume that the circuit is at steady state before $t = 0$:



$$\text{KVL}: \ 12i_x + 3(3\,i_x) + 38.5 = 0 \ \Rightarrow \ i_x = -1.8\bar{3}\ \text{A}$$

Then $v_c(0^-) = -12\,i_x = 22\ \text{V} = v_c(0^+)$

After $t = 0$:



$$\text{KVL}: \ 12i_x(t) - 8e^{-5t} + v_c(t) = 0$$

$$\text{KCL}: -i_x(t) - 2i_x(t) + (1/36)\frac{dv_c(t)}{dt} = 0 \ \Rightarrow \ i_x(t) = \frac{1}{108}\frac{dv_c(t)}{dt}$$

$$\therefore \ 12\left[\frac{1}{108}\frac{dv_c(t)}{dt}\right] - 8e^{-5t} + v_c(t) = 0$$

$$\frac{dv_c(t)}{dt} + 9v_c(t) = 72e^{-5t} \ \Rightarrow \ v_{cn}(t) = Ae^{-9t}$$

Try $v_{cf}(t) = Be^{-5t}$ & substitute into the differential equation $\Rightarrow B = 18$

$$\therefore \ v_c(t) = Ae^{-9t} + 18\,e^{-5t}$$

$$v_c(0) = 22 = A + 18 \ \Rightarrow \ A = 4$$

$$\therefore v_c(t) = 4e^{-9t} + 18e^{-5t}\ \text{V}$$